

Influenza viruses are surprisingly mutable and their genetic make-up often changes when the cells they infect churn out new viruses. "Looking at the statistics so far, I believe the epidemic may go on for some time," adds Reddy. "It's highly transmissible, but in its present state the number of deaths is not very large, proportional to the number of infected." But the medical community is watching H1N1 carefully for "drifts and shifts" (minor mutation vs formation of new, virulent strains).

Although much remains unknown, experts say the outbreak has exposed several weaknesses in the world's ability to respond to the sudden emergence of a widespread illness: that despite years of planning it is evident that the infrastructure of the health departments in many countries is inadequate to deal with the sudden appearance of a new strain of influenza; that there is still no explanation for the broad spectrum of illnesses—from few or no symptoms to rapid development

consulting agency, estimate that if vast numbers are infected and mortality rates rise, then the GDP growth can be lowered by as much as 3 per cent in the affected countries.

The Indian Government now has to beef up surveillance, increase the number of testing centres, stock up on antivirals, train doctors especially in towns and districts. There is a worldwide race to produce an effective vaccine against H1N1 which India has also joined. Off Solapur highway, cheek-by-jowl with the Poonawalla stud farm in Hadapsar, Pune, sprawls the verdant campus of the Serum Institute of India—the country's largest producer of life-saving vaccines. There's a buzz over an isolated, three-storied building—the flu central, or Pandemic Influenza Building, as they call it. "Of the select few organisations in the world to have been commissioned to develop the H1N1 vaccine by WHO, Serum is one," says Dr Rajeev Dhere, the man in charge of vaccines.

Behind closed doors, 50 scientists—goggled, gowned and gloved for safety—are brewing a special "concoction". The centrifuges whirl, microscopes move in slow motion and tens of thousands of embryonated chicken eggs turn at an angle of 45° in the incubator. "It's a major challenge and we are gearing ourselves up with every effort and resources available at our disposal to develop a vaccine as soon as possible," says Serum owner, Dr Cyrus Poonawalla.

Right now the Serum team is gearing up to check the safety and efficacy of the vaccine they have brewed on humans. "If it's successful, we will have the technology to produce a vaccine which can be pressed into service for commercial use," he adds. A welcome note of promise in a scenario where no one knows which way the battle against the bug will turn. Fingers crossed!

with Uday Mahurkar and
Amarnath K. Menon

A MOTHER AND CHILD
WAIT ANXIOUSLY AT
RML HOSPITAL IN DELHI



SUBIR HALDER

Pandemics How H1N1 stacks up

1918-20	1957-60	2009-?
SPANISH FLU	ASIAN FLU	SWINE FLU
Attack rate: 2.8%	Attack rate: 1.2%	Attack rate: 0.8-0.9%
Death rate: 4.4%	Death rate: 0.03%	Death rate: 0.8%
Age cluster: 20-40	Age cluster: Below 30	Age cluster: 15-44

Figures are a percentage of the population

If the latter happens, as has been seen in Spanish flu and the Asian flu, the second upsurge won't be easy. "We haven't found any mutation or change in the virus from different samples gathered so far," says Dr Akhilesh C. Mishra, director, National Institute of Virology, Pune. "It is the same as found in Mexico and elsewhere."

of complications and death; the presence of diarrhoea as a symptom in influenza; how it can become life-threatening in just a few hours among patients who had shown only mild symptoms; or why obesity seems to increase the risk for complication?

The burden on the Government is believed to be primarily economic. Oxford Economics, an economics